# Present Bias

Maciej Szkróbka

Based on Anujit Chakraborty (2021)

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# Agenda

- We formally define what present bias is and provide some empirical evidence of its existence.
- We present a weakening of the stationarity axiom: the Weak Present Bias axiom, which can accomodate present-biased choice reveals.
- We characterize the general class of utility functions that is consistent with the Weak Present Bias axiom both for single rewards, finite consumption streams and simple risky prospects.

# Definition

**Present bias** is the inclination to prefer smaller reward to a larger later reward, but reversing this preference when both rewards are equally delayed.

Existence of the present bias violates the axiom of **stationarity**, which holds when the relative preference for early versus later rewards depends only on the temporal distance between them.

# Examples

**Example 1.**

A: 100$ today vs. B: 110$ in a week
C: 100$ in 4 weeks vs. D: 110$ in 5 weeks

**Example 2.**

A: Juice now vs. B: Twice the amount of juice in 5 min.
C: Juice in 20 min. vs. D: Twice the amount of juice in 25 min.

# Examples

**Example 1.**

**A: 100$ today** vs. B: 110$ in a week
C: 100$ in 4 weeks vs. **D: 110$ in 5 weeks**

**Example 2.**

**A: Juice now (60%)** vs. B: Twice the amount of juice in 5 min.
C: Juice in 20 min. vs. **D: Twice the amount of juice in 25 min. (70%)**

# Notation

- The set of alternatives is the set of pairs $(x, t) \in \mathbb{X} \times \mathbb{T}$, where $x$ denoted the reward and $t$ - time.
- $\mathbb{X} = [0, M]$, where $M > 0$
- $\mathbb{T} = \mathbb{N}$ or $\mathbb{T} = [0, +\infty)$
- The decision maker has preferences $\succeq$ on this set of alternatives.

# Axioms

**A0:** $\succeq$ is complete and transitive.

**A1 (Continuity):** $\succeq$ is continous.

**A2 (Discounting):** For $t, s \in \mathbb{T}$, if $t > s$, then $(x, s) \succ (x, t)$ for $x > 0$ and $(x, s) \sim (x, t)$ for $x = 0$. For all $y > x > 0$, there exists $t \in \mathbb{T}$ such that $(x, 0) \succeq (y, t)$.

**A3 (Monotonicity):** For all $t \in \mathbb{T}$, if $x > y$, then $(x, t) \succ (y, t)$.

# Weak Present Bias Axiom

**A4 (Weak Present Bias Axiom):** If $(y, t) \succeq (x, 0)$, then $(y, t + t_1) \succeq (x, t_1)$ for all $x, y \in \mathbb{X}$ and $t, t_1 \in \mathbb{T}$.

For comparison, the stationarity axiom (used in characterizing the exponential discounting) is stated below.

*Stationarity:* $(y, t_1) \succeq (x, t_2)$ if and only if, $(y, t + t_1) \succeq (x, t + t_2)$ for all $t, t_1, t_2 \in \mathbb{T}$.

# Utility representation

**Theorem 1.** The following two statements are equivalent:

- The relation $\succeq$ defined on $\mathbb{X} \times \mathbb{T}$ satisfies A0-A4.
- For any $\delta \in (0, 1)$, there exists a set $\mathcal{U}_\delta$ of monotonically increasing continous functions $u : \mathbb{X} \to [0, 1]$ such that

$$F(x, t) = \min_{u \in \mathcal{U}_\delta} u^{-1}\Big(\delta^t u(x)\Big)$$

  represents the relation $\succeq$. The set $\mathcal{U}_\delta$ has the following properties: $u(0) = 0$ and $u(M) = 1$ for all $u \in \mathcal{U}_\delta$. $F(x, t)$ is continous.

Note that for any alternative $(x, t)$, $u^{-1}(\delta^t u(x))$ is just its present equivalent. If $t = 0$, then trivially $u^{-1}(\delta^0 u(x)) = x$ for any $u$.

# Example

Consider $\mathcal{U}_{0.91} = \{u_1, u_2\}$, where

$$u_1(x) = x^{0.99}$$

$$u_2(x) = 1 - \exp(-0.00021x)$$

In case of our first example (100\$ vs. 110\$) our representation would assign the following utilities:

$$V(100, 0) = 100$$

$$V(110, 1) = \min\{100.056, 99.995\} = 99.995$$

$$V(100, 4) = \min\{68.317, 68.48\} = 68.317$$

$$V(100, 5) = \{68.32, 68.344\} = 68.32$$

Therefore we have $(100, 0) \succ (110, 1)$ and $(100, 5) \succ (100, 4)$.

# Uniqueness

**Proposition 1.** Given the axioms A0-A4, our representation result is unique in the discount function $\Delta(t) = \delta^t$ used to calculate the present value.

**Proposition 2.** For $\delta, \alpha \in (0,1)$, if $(\delta, \mathcal{U}_\delta)$ is a representation of $\succeq$, then so is $(\alpha, \mathcal{F}_\alpha)$, where the set $\mathcal{F}_\alpha$ is constructed by the functions $v = u^{\frac{\ln \alpha}{\ln \delta}}$ for $u \in \mathcal{U}_\delta$.

# Proclivity for immediate gratification

**Definition 1.** We say that $\succeq_2$ has lower proclivity for immediate gratification than $\succeq_1$ if for all $x, y \in \mathbb{X}$ and $t \in \mathbb{T}$:

$$(x, t) \succeq_1 (y, 0) \implies (x, t) \succeq_2 (y, 0)$$

**Proposition 3.** Let $\succeq_1, \succ_2$ be two binary relations which allow for the minimum representation w.r.t. sets $\mathcal{U}_{\delta,1}$ and $\mathcal{U}_{\delta,2}$ respectively, and $\delta \in (0, 1)$. The following two statements are equivalent:

- $\succeq_2$ has a lower proclivity for immediate gratification than $\succeq_1$
- Both $\mathcal{U}_{\delta,1}$ and $\mathcal{U}_{\delta,1} \cup \mathcal{U}_{\delta,2}$ provide minimum representation of $\succeq_1$.

# Notation and axioms

Suppose the DM's preferences $\succeq$ are defined on $[0, M]^{T+1}$ with $T \geq 2$. Let $\mathbb{T} = \{0, 1, ..., T\}$. We impose the following conditions:

**D0:** $\succeq$ is complete and transitive.

**D1 (Continuity):** $\succeq$ is continous.

Given consumption stream $\mathbf{x} \in [0, M]^{T+1}$, $a \in [0, M]$, $i \in \mathbb{T}$, we define $(x_{-i}a)$ as the new consumption stream obtained by replacing the $i$-th component of stream $\mathbf{x}$ by $a$. We use $\mathbf{0} = (0, 0, ..., 0)$.

**D2 (Discounting):** If $0 \leq s < t \leq T$, then $\mathbf{0}_{-s}x \succeq \mathbf{0}_{-t}x$ for all $x \geq 0$, with the relation being strict if and only if $x = 0$.

# Axioms II

**D3 (Monotonicity):** For any $(x_i)_{i=0}^T$ ,$(y_i)_{i=0}^T \in [0, M]^{T+1}$ we have $(x_i)_{i=0}^T \succeq (y_i)_{i=0}^T$ if $x_t \geq y_t$ for all $0 \leq t \leq T$. The relation is strict if for at least one $t$ we have $x_t > y_t$.

**D4 (Weak Present Bias):** If $\mathbf{0}_{-t}y \succeq \mathbf{0}_{-0}x$, then $\mathbf{0}_{-(t+t_1)}y \succeq \mathbf{0}_{-t_1}x$ for all $x, y \in \mathbb{X}$ and $t, t_1 \in \mathbb{T}$.

For comparison, we state the stationarity axiom for consumption streams.

*Stationarity:* $(x_0, ...x_T) \succeq (y_0, ..., y_T)$ if and only if, $(x_1, ..., x_T, x_0) \succeq (y_1, ..., y_T, y_0)$ for all $x_i, y_i \in \mathbb{X}$ and $i \in \mathbb{T}$.

**D5 (Coordinate independence):** For all $i \in \mathbb{T}$, all $\mathbf{x}, \mathbf{y} \in [0, M]^{T+1}$ and $a, b \in \mathbb{X}$, then $(x_{-i}a) \succeq (y_{-i}a) \Leftrightarrow (x_{-i}b) \succeq (y_{-i}b)$.

# Utility representation

**Theorem 2.** The following two statements are equivalent:

- The relation $\succeq$ on $[0, M]^{T+1}$ satisfies properties D0-D5.
- For any $\delta \in (0, 1)$, there exists a set $\mathcal{U}_\delta$ of monotonically increasing continous functions and a continous strictly increasing function $V$ such that

$$F(x_0, ..., x_T) = \sum_{j=0}^{T} V\left( \min_{u \in \mathcal{U}_\delta} u^{-1}\left(\delta^j u(x_j)\right)\right)$$

  represents the relation $\succeq$. The set $\mathcal{U}_\delta$ has the following properties: $u(0) = 0$ and $u(M) = 1$ for all $u \in \mathcal{U}_\delta$. $F(\cdot)$ is continous. $V$ is unique up to affine transformation.

# Time-risk separability

TABLE I

FROM KEREN AND ROELOFSMA (1995)

| | Prospect A | Prospect B | % Chosing A | % Chosing B | N |
|---|---|---|---|---|---|
| 1 | (100,1,0) | (110,1,4) | 82% | 18% | 60 |
| 2 | (100,1,26) | (110,1,30) | 37% | 63% | 60 |
| 3 | (100,0.5,0) | (110,0.5,4) | 39% | 61% | 100 |
| 4 | (100,0.5,26) | (110,0.5,30) | 33% | 67% | 100 |

# Notation and axioms

Now the DM has preferences $\succeq$ over triplets $(x, p, t) \in \mathbb{X} \times [0, 1] \times \mathbb{T}$, which describe the prospect of receiving $x$ with probability $p$ in time $t$. We assume $\mathbb{T} = [0, +\infty)$. We impose the following conditions:

**B0:** $\succeq$ is complete and transitive.

**B1 (Continuity):** $\succeq$ is continous.

**B2 (Discounting):** For $t, s \in \mathbb{T}$, if $t > s$ then $(x, p, s) \succ (x, p, t)$ for $x, p > 0$ and $(x, p, s) \sim (x, p, t)$ for $x = 0$ or $p = 0$. For $y > x > 0$, there exists $T \in \mathbb{T}$ such that $(x, q, 0) \sim (y, 1, T)$.

**B3 (Prize and risk monotonicity):** For all $t \in \mathbb{T}$, $(x, p, t) \succsim (y, q, t)$ if $x \geq y$ and $p \geq q$. The preference is strict if at least one of the two following inequalities is strict.

# Axioms II

**B4 (Weak Present Bias):** If $(y, 1, t) \succsim (x, 1, 0)$, then $(y, 1, t + t_1) \succsim (x, 1, t_1)$ for all $x, y \in \mathbb{X}$, and $t, t_1 \in \mathbb{T}$.

**B5 (Probability-time tradeoff):** For all $x, y \in \mathbb{X}$, $p, q, \theta \in (0, 1]$, and $t, s, D \in \mathbb{T}$,
$(x, p\theta, t) \succsim (x, p, t + D) \Longrightarrow (y, q\theta, s) \succsim (y, q, s + D)$.

# Utility representation

**Theorem 3.** The following two statements are equivalent:

- The relation $\succeq$ satisfies B0-B5.
- For any $\delta \in (0,1)$, there exists a set $\mathcal{U}_\delta$ of monotonically increasing continous functions such that

$$F(x,p,t) = \min_{u \in \mathcal{U}_\delta} u^{-1}\Big(p\delta^t u(x)\Big)$$

represents the relation $\succeq$. The set $\mathcal{U}_\delta$ has the following properties: $u(0)=0$ and $u(M)=1$ for all $u \in \mathcal{U}_\delta$. $F(x,p,t)$ is continous.

# Summary

- We provided a definition of present bias.
- We presented the Weak Present Bias axiom and characterized the general class of utility functions consistent with it for single rewards, consumption streams and risky prospects.
- We discussed the uniqueness of the representation and the proclivity for immediate gratification.